

Apple fruit acidity is genetically diversified by natural variations in three hierarchical epistatic genes: *MdSAUR37*, *MdPP2CH* and *MdALMTII*

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SUMMARY

Many efforts have been made to map quantitative trait loci (QTLs) to facilitate practical marker-assisted selection (MAS) in plants. In the present study, using MapQTL and BSA-seq (bulk segregant analysis using next generation sequencing) with two independent pedigree-based populations, we identified four major genome-wide QTLs responsible for apple fruit acidity. Candidate genes were screened in major QTL regions, and three functional gene markers, including a non-synonymous A/G single-nucleotide polymorphism (SNP) in the coding region of *MdPP2CH*, a 36-bp insertion in the promoter of *MdSAUR37* and a previously reported SNP in *MdALMTII*, were validated to influence the malate content of apple fruits. In addition, *MdPP2CH* inactivated three vacuolar H⁺-ATPases (*MdVHA-A3*, *MdVHA-B2* and *MdVHA-D2*) and one aluminium-activated malate transporter (*MdALMTII*) via dephosphorylation and negatively influenced fruit malate accumulation. The dephosphatase activity of *MdPP2CH* was suppressed by *MdSAUR37*, which implied a higher hierarchy of genetic interaction. Therefore, the *MdSAUR37*/*MdPP2CH*/*MdALMTII* chain cascaded hierarchical epistatic genetic effects to precisely determine apple fruit malate content. An A/G SNP (−1010) on the *MdMYB44* promoter region from a major QTL (qtl08.1) was closely associated with fruit malate content. The predicted phenotype values (PPVs) were estimated using the tentative genotype values of the gene markers, and the PPVs were significantly correlated with the observed phenotype values. Our findings provide an insight into plant genome-based selection in apples and will aid in conducting research to understand the fundamental physiological basis of quantitative genetics.

Keywords: *Malus domestica* Borkh, malate, bulk segregant analysis-seq, quantitative trait locus, functional gene marker.

INTRODUCTION

In recent decades plant breeding has been greatly accelerated by marker-assisted selection (MAS). Pedigree-based quantitative trait locus (QTL) mapping and genome-wide association studies are powerful tools for mapping both qualitative and quantitative traits (Hamblin *et al.*, 2011; Longhi *et al.*, 2012; Farneti *et al.*, 2017; Ye *et al.*, 2017). Recently, bulk segregant analysis (BSA)-seq, a next-generation sequencing (NGS)-based QTL mapping strategy analysis, in which two phenotypically distinct subpopulations (bulks) (BSA extremity bulks) of recombinant progeny (segregants) are isolated from a genetic cross and genotyped, has proved successful for rapidly mapping genes in several cereal and vegetable plants (Trick *et al.*, 2012; Takagi *et al.*, 2013; Huo *et al.*, 2016; Dougherty *et al.*, 2018). By combining MapQTL and BSA-seq, an AAA ATPase gene *CsARN6.1* and the *Ef1.1* gene for waterlogging tolerance and early

flowering, respectively, in cucumber (*Cucumis sativus*), and the *ABA1/ZEP* gene for thermotolerance in lettuce (*Lactuca sativa*) were efficiently identified (Lu *et al.*, 2014; Huo *et al.*, 2016; Xu *et al.*, 2018). However, MAS is still difficult to practically apply in breeding programmes of perennial plants such as apples (*Malus domestica* Borkh.) (Kenis *et al.*, 2008; Myles, 2013; Migicovsky *et al.*, 2016; Peace, 2017).

Organic acid and soluble sugar are nutritional components and determinants of apple fruit quality. Malic acid is the most predominant organic acid and plays an important role in the flavour and palatability of apple fruit (Harker *et al.*, 2002; Zhang *et al.*, 2010; Ma *et al.*, 2015a). Previous studies have shown that the major QTLs controlling fruit acidity in apple are mapped to linkage group (LG) 16 (Liebhard *et al.*, 2003; Kenis *et al.*, 2008; Bai *et al.*, 2012; Kumar

et al., 2012; Xu et al., 2012; Ma et al., 2016), and LG 8 (Liebhard et al., 2003; Kumar et al., 2012, 2013; Zhang et al., 2012; Sun et al., 2015; Ma et al., 2016). The aluminium-activated malate transporter gene (*MdALMTII*) at the *Ma* locus on chromosome 16 with an A/G single-nucleotide polymorphism (SNP) has been reported to be associated with fruit acidity in apples (Bai et al., 2012; Khan et al., 2012; Ma et al., 2015b). However, *MdALMTII* accounted for only 7.46% of the observed phenotypic variation in malic acid content in mature apple fruits (Ma et al., 2015b). These data imply that other major gene(s) in the apple genome is (are) determinant(s) of fruit acidity. Quantitative trait loci on chromosome 8 that explained higher phenotypic variances in natural and some pedigree-based populations could also be important (Liebhard et al., 2003; Kumar et al., 2012, 2013; Zhang et al., 2012; Sun et al., 2015; Ma et al., 2016). No candidate genes have been identified that control apple fruit acidity on chromosome 8.

Several complex regulatory pathways are involved in malate metabolism in fruit cells (Sweetman et al., 2009; Etienne et al., 2013; Li et al., 2016). Vacuolar transporters such as the vacuolar pumps [vacuolar H⁺-ATPase (V-ATPase, VHA)], tonoplast dicarboxylate transporter (tDT) and especially *MdALMT* proteins play crucial roles in determining vacuolar malate transport (Emmerlich et al., 2003; Kovermann et al., 2007; Verweij et al., 2008; Schumacher and Krebs, 2010; Barbier-Brygoo et al., 2011; Meyer et al., 2011; Bai et al., 2012; Xu et al., 2012; Faraco et al., 2014; Ma et al., 2015b; Ye et al., 2017). Previous studies reported that MYB transcription factors (*MdMYB1*, *MdMYB73*) directly regulated vacuolar transporters to affect malate accumulation and vacuolar pH in apples (Hu et al., 2016, 2017).

To assess the genetic variation and determine major QTLs associated with malate content in apple fruits we performed QTL mapping based on high-density genetic linkage maps and BSA-seq on chromosomes 8 and 16 using two independent pedigree-based populations. Then, we evaluated if the predicted phenotype values of the apple fruit malate content were consistent with the observed phenotype values.

RESULTS

Stability across years and parental robustness of QTLs for fruit acidity

Four major QTLs (qtl08.1, qtl08.2, qtl08.3, qtl16.1) with stability over years and/or parental robustness were identified through MapQTL and BSA-seq approaches (Figure 1, Table 1). The fruit malate content of BSA extremity bulks is shown in Figure 1(a,b). Qtl08.1, qtl08.2 and qtl08.3 were mapped to chromosome 8 and qtl16.1 was mapped to chromosome 16. Qtl08.1 was mapped in 'Jonathan' and 'Golden Delicious' as a major QTL and in 'Zisai Pearl' and

'Red Fuji' as a minor QTL by BSA-seq. In contrast, it was detected as a minor QTL in 'Jonathan' in 2011 and 2015 by MapQTL analysis (Figure 1c–e, Table 1). Qtl08.2 was determined as a major QTL in 'Jonathan' and 'Golden Delicious' by both MapQTL and BSA-seq (Figure 1c,d, Table 1). Qtl08.3 was mapped in 'Jonathan' and 'Golden Delicious', 'Zisai Pearl' and 'Red Fuji' as a major QTL by BSA-seq and was detected as a minor QTL in 'Jonathan' in 2011 by MapQTL (Figure 1c–e, Table 1). As a major QTL, qtl16.1 was mapped in 'Red Fuji' by MapQTL and mapped in both parents 'Zisai Pearl' and 'Red Fuji' by BSA-seq (Figure 1f,g, Table 1). Other QTLs with minor effects detected by MapQTL or BSA-seq were not used in subsequent analysis due to their lower stability across years or parental robustness (Figure 1c–e,g, Table 1).

Candidate gene screening from QTLs spanning genomic regions

In all, 61, 157 and 130 genes were identified in qtl08.1, qtl08.2 and qtl08.3, respectively. Based on the annotation of the apple reference genome, the re-sequencing database of the parents and the next-generation sequencing database of segregant bulks with extremely high or low malate content, most predicted genes were culled from the list, resulting in 4, 19 and 6 genes in qtl08.1, qtl08.2 and qtl08.3, respectively (Table S1 in the online Supporting Information). These genes were amplified, sequenced and compared with the DNA template of the four parents, and 3, 16 and 4 genes were confirmed in qtl08.1, qtl08.2 and qtl08.3, respectively (Table S1). Variations in the DNA sequences of these genes of interest were re-examined using PCR or sequencing in individuals with extremely high or low malate content selected from a 'Jonathan' × 'Golden Delicious' population, and the results are presented in Table S1. Of the three genes in the qtl08.1 region, an A/G SNP –1010 bp upstream of the start codon (ATG) of the *MdMYB44* gene (MDP0000463846) exhibited the most significant segregation distortion ($P = 3.45 \times 10^{-20}$) between extremities of malate content. Therefore, this gene was considered a candidate gene in qtl08.1. Of the 19 genes in the qtl08.2 region, three genes, *MdPP2C* (MDP0000141481), *MdNADP-ME3* (MDP0000239624) and *MdMYB73* (MDP0000894463), have allele variations in 'Jonathan' with high malate content. The most distorted marker (184-bp deletion) segregation between individuals with extremely high or low malate content was observed in *MdPP2C* ($P = 1.76 \times 10^{-22}$) and thus it was selected as a candidate gene in qtl08.2. Of the four genes in qtl08.3, a 36-bp insertion 639 bp upstream of the start codon (ATG) of the *MdSAUR*-like gene (MDP0000153382) displayed highly significant distortion of segregation ($P = 1.81 \times 10^{-20}$) between extremes (high or low) of malate content, so this gene was determined to be a candidate gene in qtl08.3. In addition, *MdALMTII* (MDP0000252114) in the qtl16.1 region was

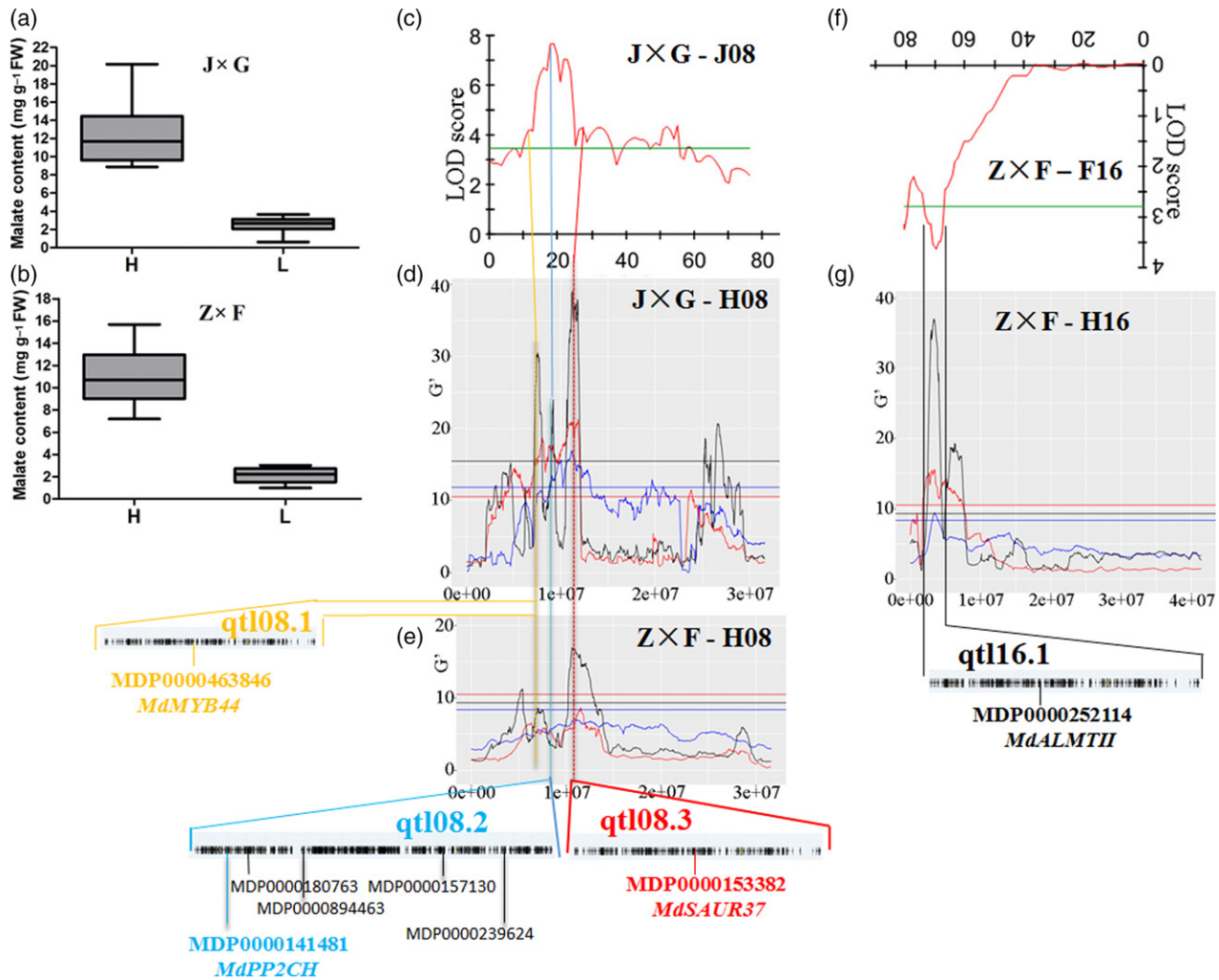


Figure 1. Quantitative trait loci (QTLs) associated with apple fruit malate content identified by MapQTL and bulk segregant analysis (BSA-seq).

(a, b) Boxplots showing the fruit malate content of BSA extremity bulks from F₁ hybrids of 'Jonathan' × 'Golden Delicious' (J×G) (a) and 'Zisai Pearl' × 'Red Fuji' (Z×F) (b). The median is indicated by the band inside the boxes. H/L, extreme high/low malate content; FW, fresh weight. (c, f) QTL charts for fruit malate content identified from J×G (c) and Z×F (f) populations by MapQTL. J08 and F16 represent chromosomes 8 and 16 of the parents 'Jonathan' and 'Red Fuji', respectively. (d, e, g) QTL charts for fruit malate content identified from J×G (d) and Z×F (e, g) populations by BSA-seq. H08 and H16 indicate the QTLs linked to heterozygous markers in both parents. The horizontal lines in the QTL charts represent the significant threshold logarithm of the odds (LOD) and G' values for MapQTL and BSA-seq analyses, respectively.

consistent with a previously reported *Ma* locus (Xu *et al.*, 2012; Ma *et al.*, 2015b), and the maximum marker segregation distortion ($P = 1.73 \times 10^{-17}$) between malate content extremities (high or low) was an A/G SNP 2579 bp downstream of the start codon (ATG) in *MdALMTII*. Therefore, it was also selected as a candidate gene.

Allelic variation in *MdALMTII* and its effect on fruit malate content

A cleaved amplified polymorphic sequences (CAPS) marker of *MdALMTII* developed by Ma *et al.* (2015b) was used to determine three genotypes (AA, AG and GG) of *MdALMTII* (Figure 2a). Among the 246 randomly selected F₁ hybrids of 'Jonathan' (AG) × 'Golden Delicious' (AG)

($P = 0.001039$) and 92 *Malus* germplasm accessions ($P = 0.008364$), the fruit malate content of hybrids with the AA genotype was significantly lower than those of AG and GG hybrids (Figure 2b,d). Notably, the fruit malate content in 123 hybrids of 'Zisai Pearl' (AG) × 'Red Fuji' (AG) displayed a clear pattern: GG > AG > AA ($P = 1.3499 \times 10^{-17}$) (Figure 2c).

Validation of allelic variations in the *MdPP2CH* gene influencing fruit malate content

Based on the apple genome database, the genomic full-length sequence of *MdPP2C* in the qtl08.2 region is 4034 bp; it comprises nine exons and eight introns coding for a 431-amino-acid (aa) protein. Phylogenetic analysis

Table 1 The quantitative trait loci (QTLs) for apple fruit malate content from MapQTL and bulk segregant analysis (BSA)-seq

Hybrid population	Mapping approaches	Year	QTLs	Parent	LOD	Exp (%)	G' value	Peak location (Mbp)	Intervals on maps (Mbp)	QTL-associated markers
J×G	BSA-seq	2016	qtl08.1	J&G			27.73	12.6	12.4–12.8	
			qtl08.2	J&G			24.91	13.6	13.4–13.8	
			qtl08.3	J&G			37.57	15.6	15.1–16.3	
			qtl08.5	J&G			16.39	23.8	23.5–24.1	
			qtl08.6	J&G			18.68	24.8	24.4–25.4	
			qtl08.7	J			13.41	8.8	8.1–9.5	
			qtl08.1	J	7.7	39.1		13.9	12.5–16.9	huC08.14919740 huC08.12762159
	MapQTL		qtl08.2							
			qtl08.3							
			qtl08.4	J	3.7	19.4		23.6	23.1–24.2	huC08.24356804 huC08.23681540
		2014	qtl08.2	J	3.3	16.5		14.1	13.5–14.2	emC08.14121899 emC08.13666872
		2015	qtl08.2	J	5.0	15.7		13.9	12.5–14.2	emC08.14121899 huC08.12762159
			qtl08.4	J	4.1	13.3		23.4	23.0–24.0	huC08.24356804 huC08.23681540
			qtl08.2	J	3.3	17.5		14.1	13.5–14.9	huC08.14919740 emC08.13666872
Z×F	BSA-seq	2016	qtl08.3	Z&F			16.93	15.6	15.1–16.4	
			qtl16.1	Z&F			37.02	1.4	0.9–2.0	
			qtl16.2	Z&F			18.77	3.8	3.1–4.5	
	MapQTL	2015	qtl16.1	F	2.9	15.8		1.5	1.1–2.1	GDsnp01588 MdSNPuiIII68
		2016	qtl16.1	F	3.5	23.9		1.3	0.9–2.2	CN948450 MdSNPui 11562

G' is a statistic accounting for BSA-seq.

J×G, 'Jonathan' × 'Golden Delicious'; Z×F, 'Zisai Pearl' × 'Red Fuji'; J, 'Jonathan'; G, 'Golden Delicious'; Z, 'Zisai Pearl'; F, 'Red Fuji'; LOD, logarithm of the odds; Exp (%), explanation of phenotypic variance (%).

showed that MdPP2C belongs to subfamily H of the PP2C family in *Arabidopsis thaliana*; thus, this gene was designated *MdPP2CH* in apples (Figure S1). A single nucleotide mutation [A/G variant at position 2831 of the mRNA and 802 of the open reading frame (ORF) in exon 6] changed an ACG codon to GCG and led to an amino acid substitution (threonine to alanine) at position 268 (T268A) (Figure S2). In addition to the A/G SNP, a 184-bp deletion (at position 2901 of the mRNA in intron 6) was confirmed (Figure 3a). Validation of the A/G SNP (2831) and 184-bp deletion (2901) through sequencing 30 *Malus* germplasm accessions provided a complete linkage between these two variations (Table S2, Figure S5a).

The 184-bp deletion of *MdPP2CH* was used as a marker to detect the segregation of the A/G SNP variant and fruit malate content (Figure 3b). Among the 246 and 123 F₁ hybrids from 'Jonathan' (AG) × 'Golden Delicious' (AG) ($P = 1.4288 \times 10^{-43}$) and 'Zisai Pearl' (GG) × 'Red Fuji' (AG) ($P = 7.1276 \times 10^{-6}$) and 92 *Malus* germplasm accessions ($P = 1.1597 \times 10^{-13}$), the fruit malate content of the GG genotype hybrids was higher than that of the AG genotype, and the lowest fruit malate content was observed in the AA genotype (Figure 3c–e).

Structure prediction of the allelic amino acid variant T268A of MdPP2CH revealed that the protein structure of the variation site adapted to two α -helices from three α -helices (Figure 3f,g). To test the presence of phosphatase activity in the two MdPP2CH variants (MdPP2CH^{268T} and MdPP2CH^{268A}) and their differences in enzymatic activity, the full-length coding sequence (ORF) of *MdPP2CH* was cloned from 'Jonathan' and 'Golden Delicious', and the GST-tagged recombinant fusion protein of the 268T and 268A variants was obtained (Figure S3). The phosphatase activity of the two recombinants was tested *in vitro* using p-nitrophenyl-phosphate (pNPP) as a substrate with an additional MgCl₂ or MnCl₂ treatment (Hayashi *et al.*, 2010). The data demonstrated that the phosphatase activity of MdPP2CH was dependent on Mn²⁺ (Figure 3h,i) and that of MdPP2CH^{268T} was significantly greater than that of MdPP2CH^{268A} in genes cloned from 'Jonathan' (J) or 'Golden Delicious' (G) whether or not Mn²⁺ was present (Figure 3h,i).

We conducted subcellular localization analysis to investigate whether intracellular compartmentalization of MdPP2CH^{268T} and MdPP2CH^{268A} protein was influenced by the single amino acid mutation T268A. Two constructs,

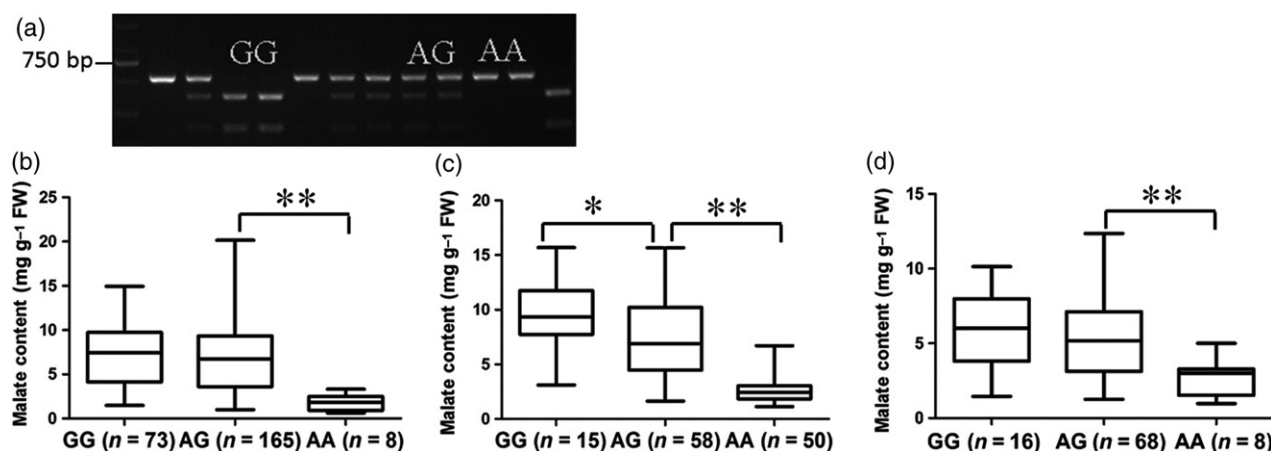


Figure 2. Association between the allelic variation of *MdALMTII* and fruit malate content.

(a) A cleaved amplified polymorphic sequence marker was used to determine the genotypes (AA, AG and GG) of *MdALMTII* and agarose gel electrophoresis was conducted on the PCR products digested with *HaeIII*.

(b–d) Association analysis between *MdALMTII* genotypes (AA, AG and GG) and malate content in 246 hybrids from 'Jonathan' × 'Golden Delicious' (b), 123 hybrids from 'Zisai Pearl' × 'Red Fuji' (c), and 92 *Malus* germplasm accessions (d). * $P < 0.05$ and ** $P < 0.01$ by Dunnett's multiple comparison. FW, fresh weight. [Colour figure can be viewed at wileyonlinelibrary.com].

35S:MdPP2CH^{268T}-GFP and 35S:MdPP2CH^{268A}-GFP, were transiently transformed into *Nicotiana benthamiana* leaf cells using *Agrobacterium* infection and into apple calli protoplasts by polyethylene glycol (PEG)-mediated transformation. The 35S:MdPP2CH^{268T}-GFP and 35S:MdPP2CH^{268A}-GFP fusion protein signals were observed in the nucleus, cytoplasm, tonoplast and cell membrane with confocal microscopy (Figure 3j,k). These results indicated that the variation between MdPP2CH^{268T} and MdPP2CH^{268A} did not alter subcellular localization.

Five constructs, including 35S:MdPP2CH^{J268T}-GFP, 35S:MdPP2CH^{J268A}-GFP, 35S:MdPP2CH^{G268T}-GFP, 35S:MdPP2CH^{G268A}-GFP and RNAi:MdPP2CH, were transformed into apple calli (from the flesh of 'Orin') to test the function of MdPP2CH (Figure 4a). The expression of *MdPP2CH* was significantly higher in overexpressing transgenic calli, and it was much lower in RNAi transgenic calli compared with wild-type (WT) calli (Figure 4c). Immunoblot assays with an anti-GFP antibody showed that MdPP2CH protein was detected in the *MdPP2CH* transgenic calli but the protein was not observable in the WT control (Figure S4). The malate content was significantly lower in 35S:MdPP2CH^{J268T}-GFP and 35S:MdPP2CH^{G268T}-GFP transgenic calli lines than in WT, 35S:MdPP2CH^{J268A}-GFP and 35S:MdPP2CH^{G268A}-GFP transgenic calli lines, whereas the malate content was greater in RNAi:MdPP2CH transgenic calli than in the WT (Figure 4b), implying that MdPP2CH negatively regulated the accumulation of fruit malate and its function was considerably reduced by T268A mutation.

Interaction between MdPP2CH and vacuolar H⁺-ATPase

Protein interactions between MdPP2CH and MdVHA-A3, MdVHA-B2, MdVHA-D2 and MdALMTII were identified by

yeast two-hybrid (Y2H) screening. Positive X- α -gal activity was observed in yeast containing AD-MdPP2CH and BD-MdVHA-A3, BD-MdVHA-B2, BD-MdVHA-D2 and BD-MdALMTII but not in yeast containing the empty vector BD (Figure 4f). These interactions were verified by bimolecular fluorescence complementation assay (BiFC). After co-transforming NE-MdPP2CH and CE-MdVHA-A3, NE-MdPP2CH and CE-MdVHA-B2, NE-MdPP2CH and CE-MdVHA-D2, or NE-MdPP2CH and CE-MdALMTII into apple calli protoplasts, a yellow fluorescence signal was observed in the vacuolar membrane and cytoplasm (Figure 4g). The MdPP2CH partially contributed to the decrease in malate content via dephosphorylation of MdVHA-A1, MdVHA-B2, MdVHA-D2 and MdALMTII, possibly leading to a decline in malate transport into vacuoles.

In addition, the V-ATPase and proton-pumping activities in 35S:MdPP2CH^{J268T}-GFP and 35S:MdPP2CH^{G268T}-GFP transgenic calli lines showed an approximate two-fold decrease compared with those in WT calli whereas there was an approximate 1.2- to 1.5-fold decrease in 35S:MdPP2CH^{J268A}-GFP and 35S:MdPP2CH^{G268A}-GFP transgenic calli (Figure 4d,e). These data indicated that MdPP2CH negatively regulated tonoplast V-ATPase activity and fruit malate content.

The expression pattern and promoter activity of *MdPP2CH*

The fruit malate content in 'Jonathan' and three F₁ hybrids with the GG genotype of *MdPP2CH* (52-049, 52-120 and 52-201) was significantly greater than that in 'Golden Delicious' and three F₁ hybrids with AA genotype (52-146, 54-047 and 55-008) at 30–150 days after full bloom (DAFB) except at 30 DAFB (Figure 5a). We did not observe robust differences in the expression of *MdPP2CH* between hybrids

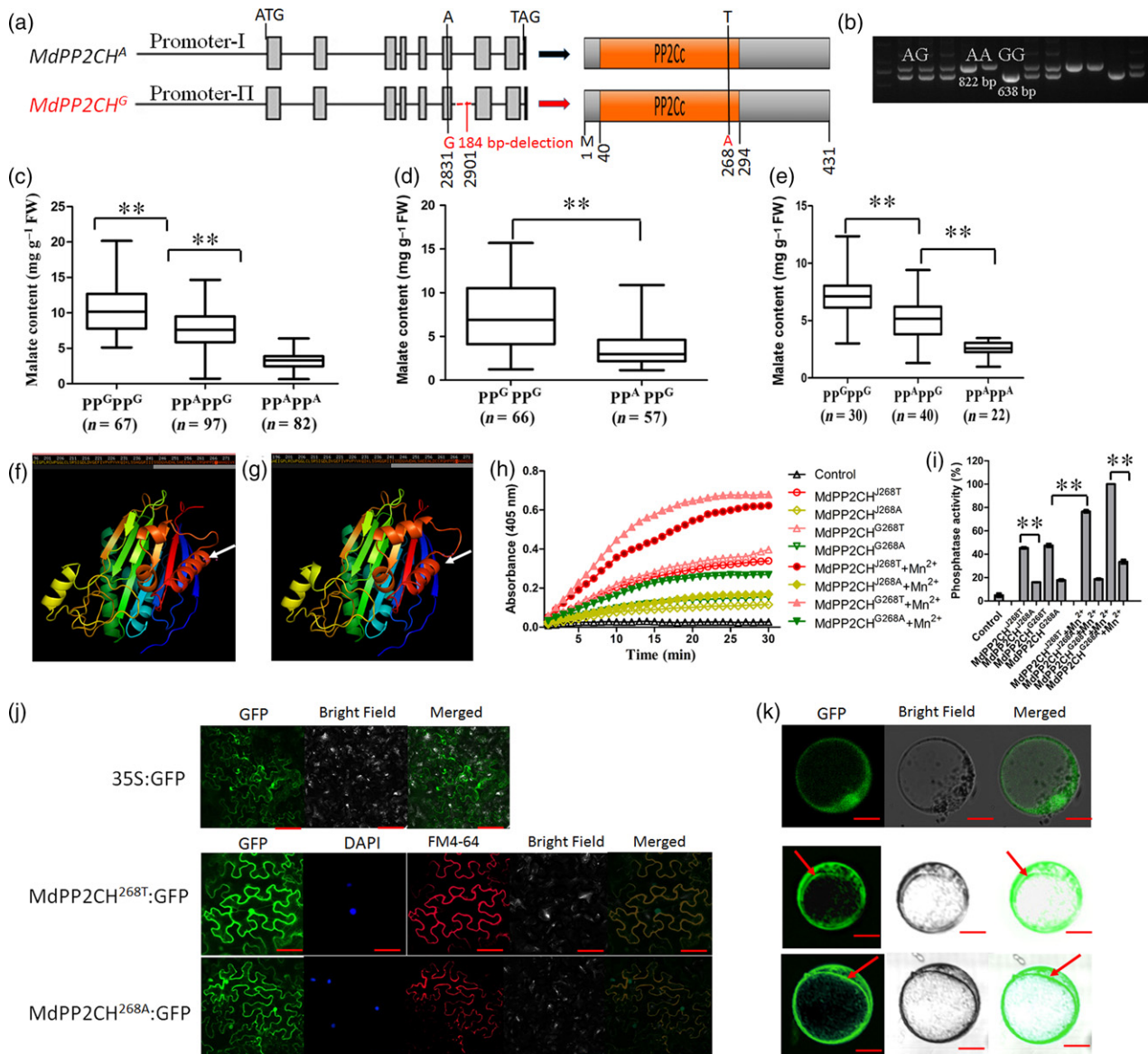


Figure 3. Identification of the allelic variations in MdPP2CH.

(a) Structural variations of the *MdPP2CH* haplotypes. Boxes represent coding sequences and thin lines between boxes represent introns. (b) The 184-bp deletion marker was used to determine the genotypes (PP^APP^A, PP^APP^G, PP^GPP^G) of *MdPP2CH* via agarose gel electrophoresis of the PCR products. (c–e) Analysis of association was conducted between the genotypes (PP^GPP^G, PP^APP^G, PP^APP^A) of *MdPP2CH* and the malate content in 246 hybrids from ‘Jonathan’ × ‘Golden Delicious’ (c), 123 hybrids from ‘Zisai Pearl’ × ‘Red Fuji’ (d) and 92 *Malus* accessions (e) (FW, fresh weight). (f, g) The three-dimensional structures of MdPP2CH^{268T} (f) and MdPP2CH^{268A} (g) were predicted using SWISS modelling. (h) Phosphatase activity was assayed and the release of *p*-nitrophenol was monitored colorimetrically at 405 nm every min for 30 min. (i) A comparison of phosphatase activity between MdPP2CH^{268T} and MdPP2CH^{268A}. (j, k) Subcellular co-localization of transiently expressed MdPP2CH^{268T}-GFP and MdPP2CH^{268A}-GFP fusion protein in *Nicotiana benthamiana* leaves (j) and apple calli protoplasts (k). 4',6-Diamidino-2-phenylindole was used to stain the nucleus. FM4-64 was used to stain the tonoplasts. Arrows highlight the position of vacuoles exhibiting green fluorescence. The fluorescent images were obtained using confocal microscopy. Scale bars = 50 μm. ***P* < 0.01 by Dunnett's multiple comparison test.

and the parents with high or low malate content (Figure 5b), and the correlation between malate content and the expression of *MdPP2CH* in ripened apples of the 30 F₁ hybrids of ‘Jonathan’ × ‘Golden Delicious’ with different A/G variations was not significant ($R^2 = 0.0407$, $P > 0.5$) (Figure 5c).

We constructed a 2672-bp promoter of *MdPP2CH* with a GUS reporter as MdPP2CHpro-JA:GUS (linkage with A)

and MdPP2CHpro-JG:GUS (linkage with G) from ‘Jonathan’, and MdPP2CHpro-GA:GUS (linkage with A) and MdPP2CHpro-GG:GUS (linkage with G) from ‘Golden Delicious’ (Figure S5b). The four constructs were transiently expressed in *N. benthamiana* leaf cells and apple calli protoplasts. The GUS reporter exhibited no obvious differences among the four transformants of the four *MdPP2CH* promoters (Figure 5d). Therefore, the expression

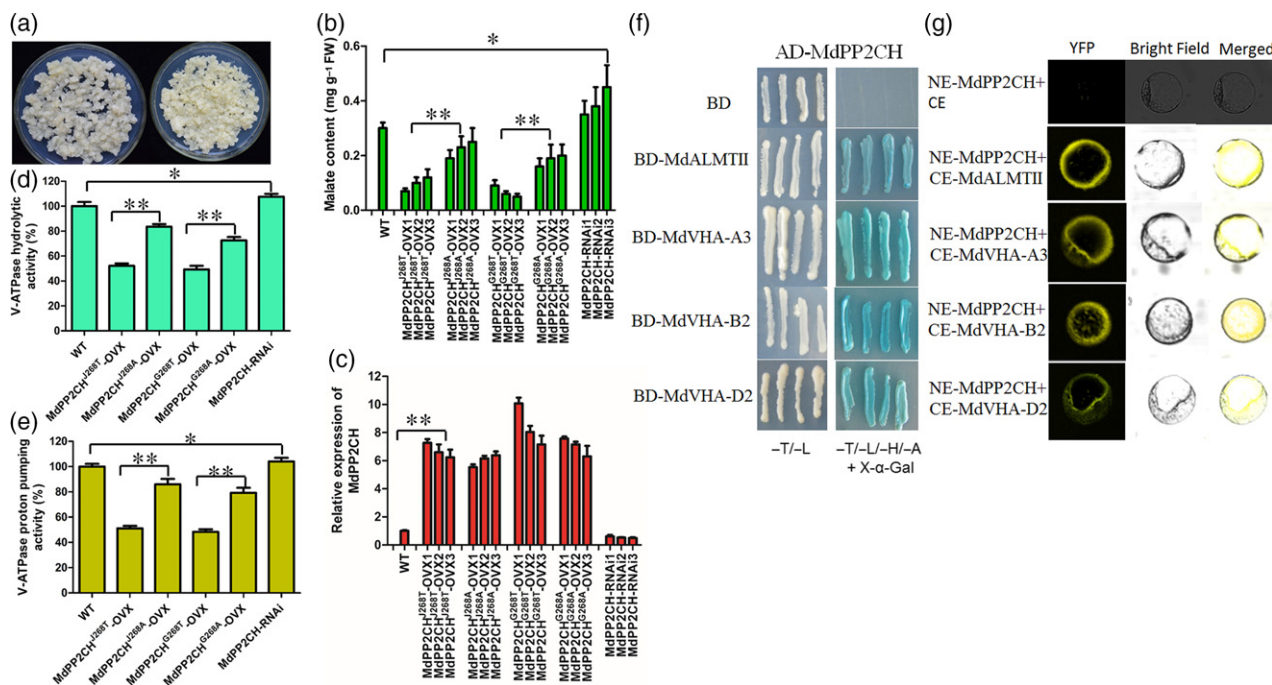


Figure 4. MdPP2CH can interact with MdALMTII, MdVHA-A3, MdVHA-B2 and MdVHA-D2 to negatively modulate apple fruit malate accumulation (a) Transgenic calli lines were obtained by transforming 35S:MdPP2CH^{J268T}-GFP, 35S:MdPP2CH^{J268A}-GFP, 35S:MdPP2CH^{G268T}-GFP, 35S:MdPP2CH^{G268A}-GFP and RNAi:MdPP2CH. (b) Malate content in wild type (WT) and MdPP2CH transgenic apple calli (FW, fresh weight). (c) The transcripts of MdPP2CH in WT and transgenic apple calli lines. (d) ATP hydrolysis activity of V-ATPase in WT and MdPP2CH transgenic calli lines. (e) Proton-pumping activity of V-ATPase in WT and MdPP2CH transgenic calli lines. (f) MdPP2CH interacted with MdALMTII, MdVHA-A3, MdVHA-B2 and MdVHA-D2 in a yeast two-hybrid assay. (g) Bimolecular fluorescence complementation with MdPP2CH and MdALMTII, MdVHA-A3, MdVHA-B2 and MdVHA-D2. The indicated split YFP constructs were transiently co-expressed in protoplasts of apple calli and the fluorescent images were obtained using confocal microscopy. Bars = 50 mm. * $P < 0.05$ and ** $P < 0.01$ by Dunnett's multiple comparison.

level of *MdPP2CH* did not contribute to the variation of fruit malate content.

The involvement of allelic variations of *MdSAUR*-like in controlling fruit malate content

The full-length sequence of the candidate gene *MdSAUR*-like in qtl08.3 is 522 bp (encoding a 174-aa protein); it does not contain an intron. Phylogenetic analysis showed that *MdSAUR*-like belongs to subfamily 37 of the SAUR family in *A. thaliana*, so it was designated *MdSAUR37* in apple (Figure S6). A 36-bp insertion (–639 bp upstream of the ATG codon) in the promoter region, a SNP (G/A variant at position –17) and a synonymous SNP in exon (G/A variant at position 474 of the ORF) were detected in the *MdSAUR37* gene (Figure 6a).

The 36-bp insertion in the *MdSAUR37* promoter region was developed as a marker. S^{P136} (I) and S^P (ii) represent the presence or absence of the 36-bp insertion, respectively (Figure 6b). Among the 246 and 123 F₁ hybrids from 'Jonathan' (li) × 'Golden Delicious' (li) ($P = 2.3518 \times 10^{-39}$) and 'Zisai Pearl' (li) × 'Red Fuji' (li) ($P = 2.1823 \times 10^{-5}$) populations, and 92 *Malus* germplasm accessions ($P = 4.9369 \times 10^{-16}$), the fruit malate content of hybrids with genotype II was greater than those with genotype I,

with the lowest fruit malate content being observed in genotype ii (Figure 6c–e).

Expression of *MdSAUR37* in F₁ hybrids with a high fruit malate content was significantly higher than in hybrids with a low fruit malate phenotype at all fruit developmental stages (Figure 6f). Its expression also exhibited a close correlation ($R^2 = 0.7792$, $P < 0.001$) with fruit malate content in different genotypes in ripened fruit (Figure 6g).

To determine if the 36-bp insertion in the *MdSAUR37* promoter affected its expression, a 1400-bp promoter of *MdSAUR37* was cloned from 'Jonathan' and 'Golden Delicious' (MdSAUR37pro-JI and MdSAUR37pro-GI contained the 36-bp insertion whereas MdSAUR37pro-J and MdSAUR37pro-G did not contain the insertion) (Figure S7). The sequences of MdSAUR37pro-J and MdSAUR37pro-G were the same. Six constructs, MdSAUR37pro-JI:GUS, MdSAUR37pro-JI-36:GUS, MdSAUR37pro-GI:GUS, MdSAUR37pro-GI-36:GUS, MdSAUR37pro-J:GUS and MdSAUR37pro-J + 36:GUS, were generated and transformed into *N. benthamiana* leaf cells and apple calli protoplasts (Figure 6h). The GUS reporter expressions exhibited a two- to three-fold increase driven by the 36-bp insertion promoter (Figure 6i). These results suggest that the 36-bp insertion enhanced basal promoter activity.

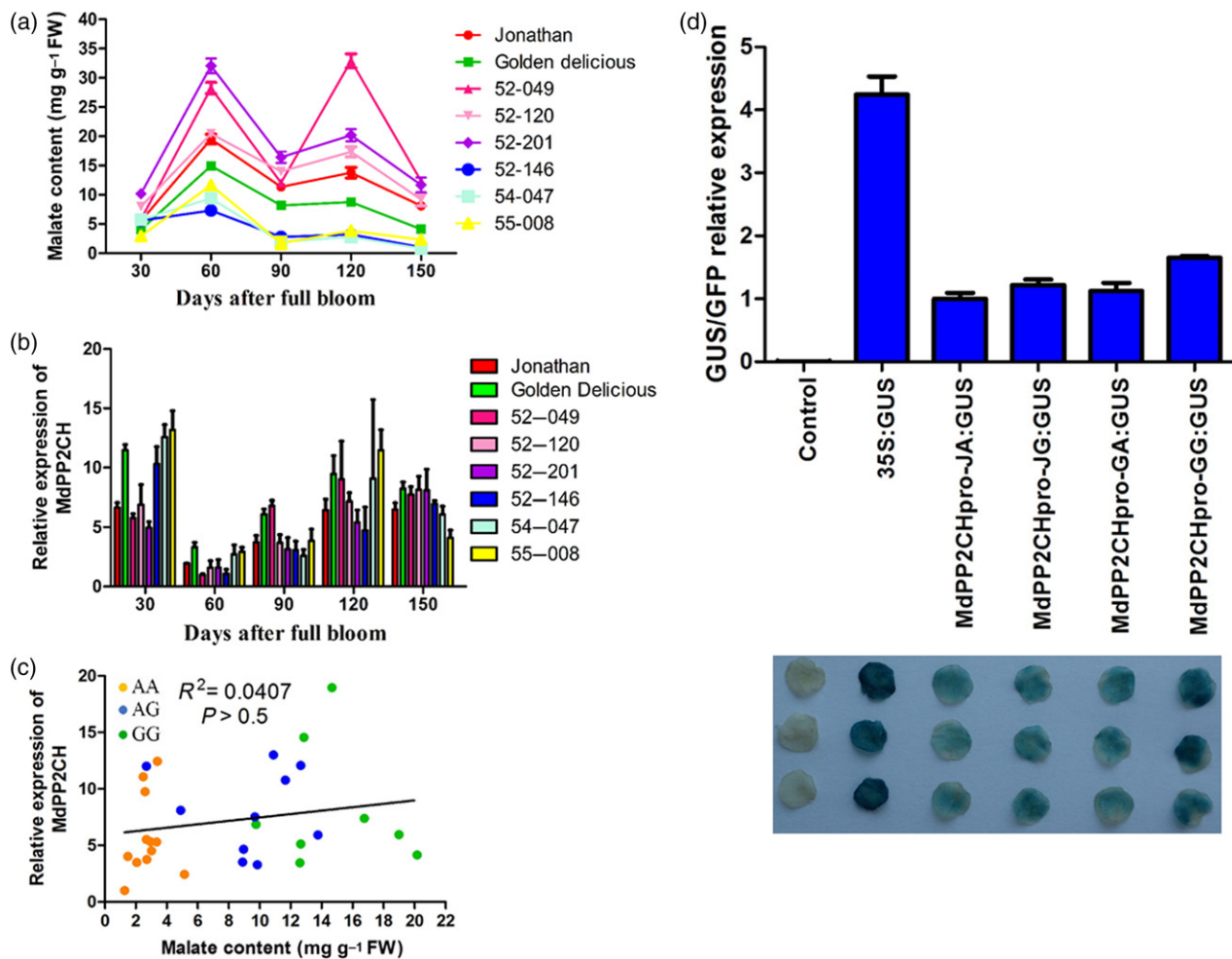


Figure 5. The expression pattern and promoter activity of *MdPP2CH*.

(a) Changes of malate content during fruit development of different apple varieties (FW, fresh weight). (b) Expression profiles of the *MdPP2CH* gene during fruit development of different apple varieties. (c) Correlation analysis between relative expression levels of *MdPP2CH* and fruit malate content. (d) Transient expression analysis of *MdPP2CH* promoters in apple calli protoplasts. GUS expression was measured relative to that of GFP driven by the constitutive 35S promoter. The error bars in (d) indicate the standard deviation of the three biological replicates. The transgenic tobacco leaves were dyed using a β -Galactosidase Reporter Gene Staining Kit.

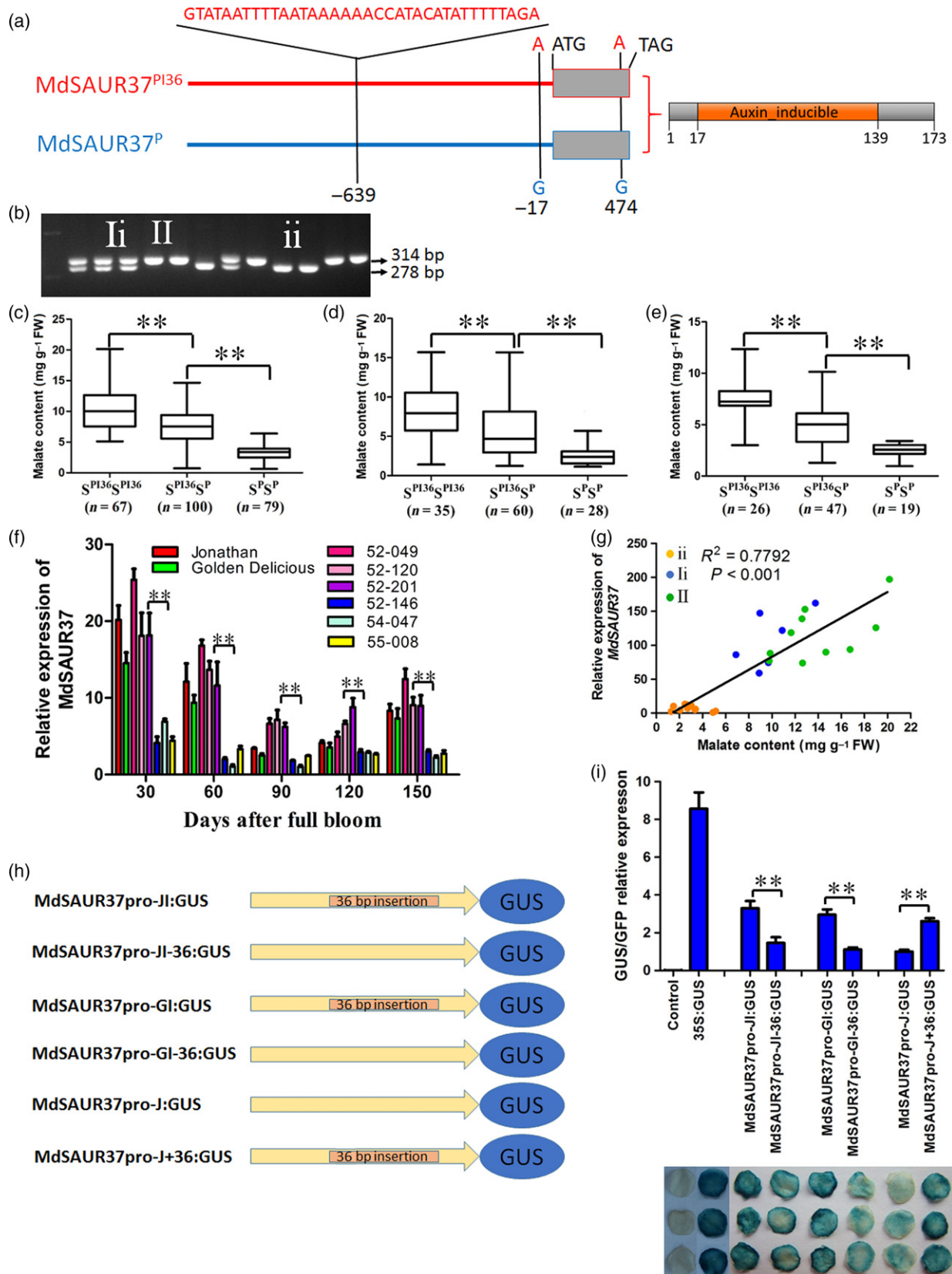
MdSAUR37 contributed to high fruit malate content by negatively regulating *MdPP2CH*

Subcellular localization analysis of MdSAUR37 demonstrated that the fusion protein signals were identified in the nucleus and cytosol (Figure 7a).

A Y2H screening was performed to test the interaction between MdSAUR37 and MdPP2CH. Positive X- α -gal activity was observed in yeast containing AD-MdPP2CH and BD-MdSAUR37, indicating that MdSAUR37 interacted with MdPP2CH (Figure 7b). The subsequent BiFC assay

Figure 6. Identification of the 36-bp insertion in the promoter of *MdSAUR37*.

(a) Structural variations of the *MdSAUR37* haplotypes. Boxes represent the coding sequences. (b) The 36-bp insertion marker was used to determine the genotypes ($S^{PI36}S^{PI36}$, $S^{PI36}S^P$, $S^P S^P$) of *MdSAUR37* by agarose gel electrophoresis of the PCR products. (c–e) Analysis of association between the genotypes ($S^{PI36}S^{PI36}$, $S^{PI36}S^P$, $S^P S^P$) of *MdSAUR37* and malate content in 246 hybrids from 'Jonathan' $\times$ 'Golden Delicious' (c), 123 hybrids from 'Zisai Pearl' $\times$ 'Red Fuji' (d) and 92 *Malus* accessions (e) (FW, fresh weight). (f) Expression profiles of the *MdSAUR37* gene during fruit development of different apple varieties. (g) Correlation analysis between relative expression levels of *MdSAUR37* and fruit malate content. (h) Schematic of *MdSAUR37* promoters with or without the 36-bp insertion. (i) Transient expression analysis of *MdSAUR37* promoters with or without the 36-bp insertion in apple calli protoplasts. GUS expression was measured relative to that of GFP driven by the constitutive 35S promoter. The error bars in (i) indicate the SD of three biological replicates. The transgenic tobacco leaves were dyed using a β -Galactosidase Reporter Gene Staining Kit. $^{**}P < 0.01$ by Dunnett's multiple comparison.



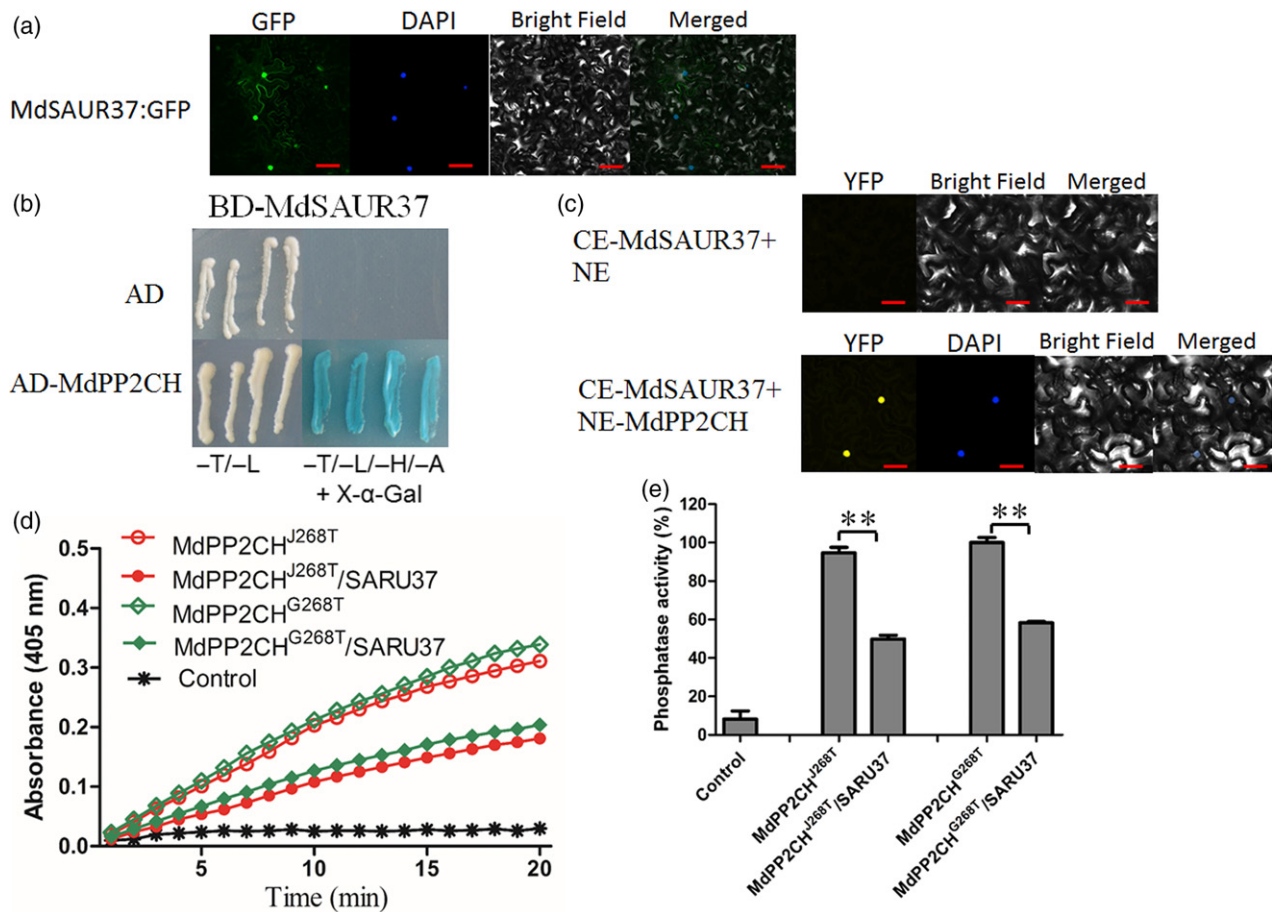


Figure 7. MdSAUR37 interacts with MdPP2CH and inhibits MdPP2CH activity.

(a) Subcellular localization of transiently expressed MdSAUR37-GFP fusion protein in *Nicotiana benthamiana* leaves. Scale bars = 50 μm.

(b) MdSAUR37 interacted with MdPP2CH in a yeast two-hybrid assay.

(c) Bimolecular fluorescence complementation with MdSAUR37 and MdPP2CH. The indicated split YFP constructs were transiently co-expressed in *N. benthamiana* leaves and fluorescent images were obtained using confocal microscopy. Bars = 50 μm. FW, fresh weight.

(d) *p*-Nitrophenyl-phosphate phosphatase assays containing GST-MdPP2CH with or without GST-MdSAUR37.

(e) Phosphatase activity of MdPP2CH when treated with MdSAUR37 protein. The error bars in (e) indicate the SD of three biological replicates. ** $P < 0.01$ by Dunnett's multiple comparison.

confirmed the interaction between MdPP2CH and MdSAUR37 and a yellow fluorescence signal was detected in the nucleus and cytosol by confocal microscopy (Figure 7c).

We examined if MdSAUR37 binding could influence MdPP2CH activity. *In vitro* phosphatase assays indicated that recombinant bacterial GST-tagged fusion proteins of MdSAUR37 (Figure S8) and MdPP2CH (Figure S3) were expressed and the activity of GST-MdPP2CH was reduced by the addition of GST-MdSAUR37 (Figure 7d). MdPP2CH activity was reduced by approximately 35% by the addition of MdSAUR37 (Figure 7e).

Validation of allelic variations in *MdMYB44* affecting fruit malate content

The candidate gene *MdMYB44* in qtl08.1 was also cloned and sequenced. The full length of the *MdMYB44* coding

sequence (CDS) is 960 bp with no intron and it encodes a 320-aa protein. In the promoter region, a single nucleotide deletion T (T/–, –536 upstream of the ATG codon), an SNP (A/G variant –1010) and a SNP (T/A variant –1647) were detected in *MdMYB44* (Figure 8a). The A/G SNP (–1010) and the T/– SNP (–536) were validated as complete linkages by sequencing *Malus* 30 germplasm accessions (Table S3).

The A/G SNP (–1010) in the *MdMYB44* promoter region was sequenced to test the association of this SNP genotype (AA, AG, GG) with the malate content. Among the 246 and 123 F₁ hybrids from 'Jonathan' (AG) × 'Golden Delicious' (AG) ($P = 6.3894 \times 10^{-36}$) and 'Zisai Pearl' (GG) × 'Red Fuji' (AG) ($P = 5.5797 \times 10^{-8}$) and 92 *Malus* germplasm accessions ($P = 6.3898 \times 10^{-6}$), the fruit malate content of individuals with the GG genotype were relatively higher than that of those with the AG genotype, with

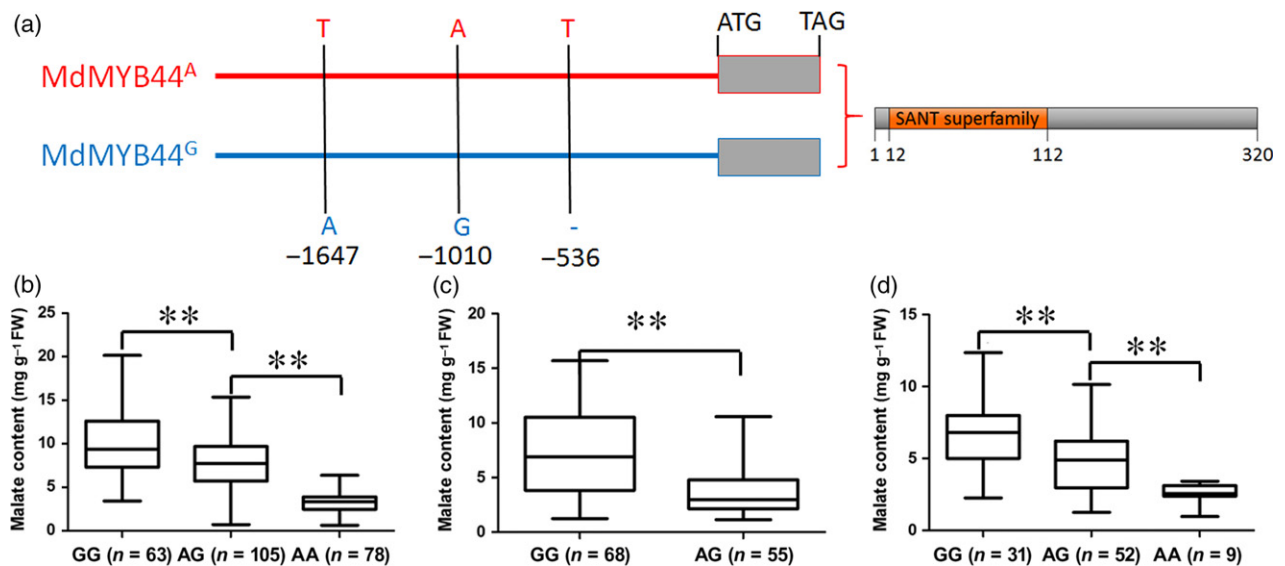


Figure 8. Association between the A/G single-nucleotide polymorphism of *MdMYB44* and apple fruit malate content.

(a) Structural variations of the *MdMYB44* haplotypes. Boxes represent the coding sequences.

(b–d) Analysis of association between the genotypes (AA, AG, GG) of *MdMYB44* and malate content was determined in 246 hybrids from 'Jonathan' × 'Golden Delicious' (b), 123 hybrids from 'Zisai Pearl' × 'Red Fuji' (c) and 92 *Malus* accessions (d). ** $P < 0.01$ by Dunnett's multiple comparison. FW, fresh weight. [Colour figure can be viewed at wileyonlinelibrary.com].

the lowest fruit malate content being found in the AA genotype (Figure 8b–d).

Analysis of genetic linkages between *MdSAUR37*, *MdPP2CH* and *MdMYB44* genotypes

To test the linkage between the genotypes of *MdSAUR37*, *MdPP2CH*, *MdALMTII* and *MdMYB44* variants, 246 and 123 randomly chosen hybrids from 'Jonathan' × 'Golden Delicious' and 'Zisai Pearl' × 'Red Fuji', respectively, and 92 *Malus* germplasm accessions were genotyped. As shown in Table S4, the genotypes of *MdALMTII* were independent of the genotypes of any other genes, but the linkage between the i haplotype of *MdSAUR37*, A of *MdPP2CH* and A of *MdMYB44* was obvious; the recombination rates were estimated at 5.69% (14/246), 13.41% (33/246) and 17.89% (44/246) between *MdMYB44*/*MdPP2CH*, *MdPP2CH*/*MdSAUR37* and *MdMYB44*/*MdSAUR37*, respectively (Table S5). Similar recombination rates were estimated in hybrids of 'Zisai Pearl' × 'Red Fuji' and *Malus* germplasm accessions (Tables S4 and S5).

Prediction of fruit acidity phenotype values with genotypes of *MdSAUR37*, *MdPP2CH*, *MdALMTII* and *MdMYB44*

Close associations between fruit malate content and genotypes of *MdSAUR37*/*MdPP2CH*/*MdALMTII*/*MdMYB44* variations in 246 F₁ hybrids from 'Jonathan' × 'Golden Delicious' ($P = 8.7695 \times 10^{-42}$), 123 F₁ hybrids from 'Zisai Pearl' × 'Red Fuji' ($P = 1.1710 \times 10^{-19}$) and 92 *Malus* germplasm accessions ($P = 3.4179 \times 10^{-11}$) are

schematically shown in Figure 9(a–c) and Table S6. The ratios between allelic variations were estimated using observed phenotype values and the genotype values (GVs) were tentatively assigned to the genes *MdSAUR37* (ii = 1, li = 2 and ll = 3), *MdPP2CH* (AA = 1, AG = 2 and GG = 3), *MdALMTII* (AA = 0.26, AG = 1 and GG = 1.1) and *MdMYB44* (AA = 1, AG = 1.75 and GG = 2) (Table S7). We proposed that predicted phenotype values (PPVs) should be the summation of the multiplication of GV of *MdSAUR37*, *MdPP2CH*, and *MdALMTII* and the GV of *MdMYB44* because a hierarchically regulatory relationship of the first three genes and the independence of the variation in *MdMYB44* from fruit malate content were shown. Therefore, the PPV should be: $PPV = GV(MdSAUR37) \times GV(MdPP2CH) \times GV(MdALMTII) + GV(MdMYB44)$ (Table S8). The PPVs were significantly correlated with the observed fruit malate content for the phenotype, with regression coefficients from 0.7646 to 1.011 and R^2 values larger than 0.6606 ($P < 0.001$) (Figure 9d–f).

DISCUSSION

An ideal MAS practice in plants relies on successful identification of almost all QTLs and precise estimation of their effects on the phenotypic variance; therefore genome-based selection (GBS) was suggested by Meuwissen *et al.* (2001). Previous studies reported that QTLs varied among pedigree-based populations, year of plant phenotyping and mapping approaches (Kenis *et al.*, 2008; Kumar *et al.*, 2012, 2013; Xu *et al.*, 2012; Sun *et al.*, 2015). In our study, qtl08.2 was identified as a major QTL in

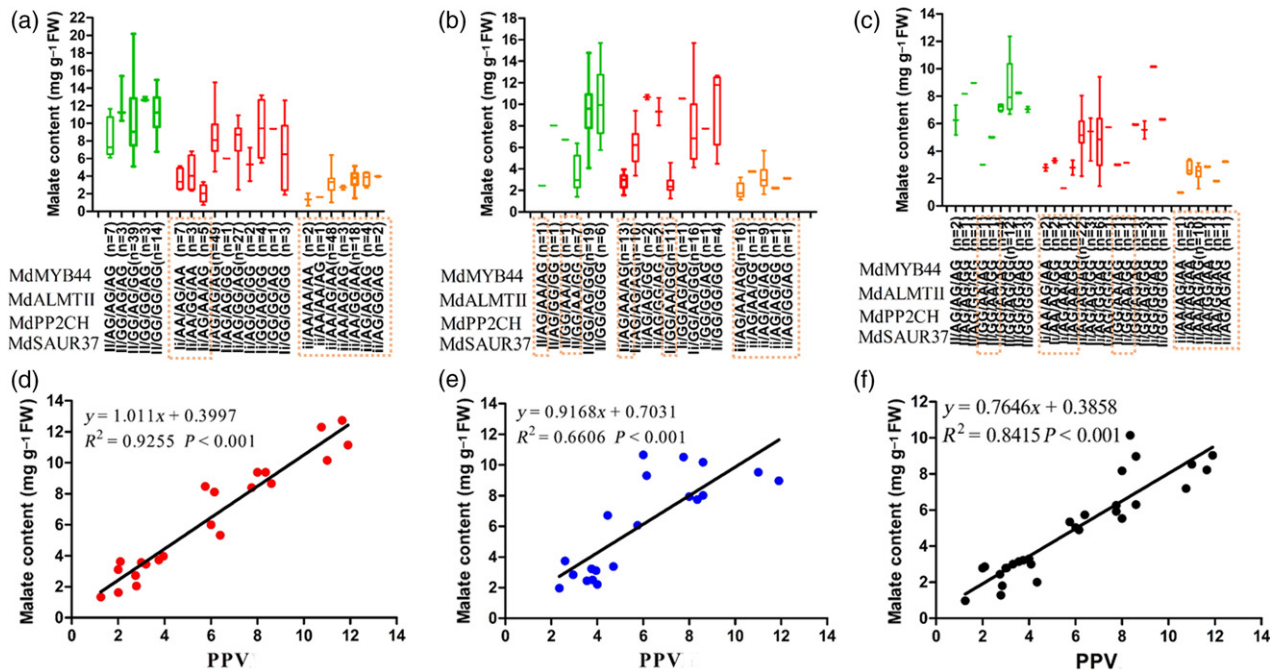


Figure 9. Association between genotypes of *MdSAUR37*/*MdPP2CH*/*MdALMTII*/*MdMYB44* and fruit malate content.

(a–c) Association analysis of *MdSAUR37*/*MdPP2CH*/*MdALMTII*/*MdMYB44* genotypes and fruit malate content in 246 hybrids from 'Jonathan' × 'Golden Delicious' (a), 123 hybrids from 'Zisai Pearl' × 'Red Fuji' (b) and 92 *Malus* accessions (c) (FW, fresh weight). (d–f) Linear regression analysis between predicted phenotype values (PPVs) and observed phenotype values of fruit malate content in 246 hybrids from 'Jonathan' × 'Golden Delicious' (d), 123 hybrids from 'Zisai Pearl' × 'Red Fuji' (e) and 92 *Malus* accessions (f). [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)].

'Jonathan' × 'Golden Delicious' but not in 'Zisai Pearl' × 'Red Fuji' because the A/G SNP on *MdPP2CH* was heterozygous in three parents 'Jonathan', 'Golden Delicious' and 'Red Fuji', whereas it was homologous (G/G) in 'Zisai Pearl'. A similar phenomenon was observed in qtl08.1. Although the *MdALMTII* (A/G) genotype was heterozygous in all four parents and the major qtl16.1 was detected in 'Zisai Pearl' × 'Red Fuji', it was undetectable in 'Jonathan' × 'Golden Delicious' using BSA-seq or MapQTL approaches (Figure 1, Table 1) (Zhang *et al.*, 2012; Sun *et al.*, 2015), possibly due to fewer AA genotypes (8/246) than theoretically expected from the segregation ratio (1/4) observed in the 'Jonathan' × 'Golden Delicious' population. Therefore, future efforts should study several independent and/or natural segregating populations to achieve GBS of quantitative traits in apple fruit acidity (Kumar *et al.*, 2012; Kim *et al.*, 2015; Ye *et al.*, 2017).

Genetic epistasis can be explained by upstream regulation of genes in physiology. The *ALMT* genes have been confirmed to correlate with fruit acidity in apples and tomatoes (Bai *et al.*, 2012; Khan *et al.*, 2012; Xu *et al.*, 2012; Ma *et al.*, 2015b, 2016; Ye *et al.*, 2017). *MdALMTII* in apple was suggested to function as a channel for increasing the influx of malate into the tonoplast and an A/G SNP on *MdALMTII* led to a stop-gain variant of the protein and caused loss of function (Ma *et al.*, 2015b). Previous studies suggested that H⁺-ATPase was dephosphorylated by Mg²⁺/

Mn²⁺-dependent phosphatase and PP2C-D1 negatively regulated AHA2 function (Hayashi *et al.*, 2010; Spartz *et al.*, 2014). In apples, vacuolar proton pump subunit genes, including *MdVHA-A*, *MdVHA-B*, *MdVHA-D* and *MdALMTII*, can modulate the accumulation and pH of vacuolar malate (Hu *et al.*, 2016). According to our data, the *MdALMTII* protein was inactivated by *MdPP2CH* by dephosphorylation; hence, the variation in *MdPP2CH* affected *MdALMTII* and fruit acidity control. In addition, the dephosphatase activity of *MdPP2CH* was suppressed by *MdSAUR37*, indicating a higher hierarchy of intergenetic interaction. Therefore, an *MdSAUR37*/*MdPP2CH*/*MdALMTII* regulatory chain that cascades hierarchical epistatic genetic effects on apple fruit malate content was postulated (Figures S9 and S10). Genetic epistasis is ubiquitous and is a major determinant of genetic variance in plants (Wittenburg *et al.*, 2011; Powell *et al.*, 2013; Wu *et al.*, 2013; Monahan and Kelly, 2015). We assigned GVs to the three epistatic genes, and the PPVs were estimated by multiplying the proposed GVs. To our surprise, the PPVs were remarkably consistent with the observed phenotype values. These data enlighten GBS prospects in plants and encourage further work to clarify the physiological fundamentals of quantitative genetics.

PP2C protein dephosphatases are ubiquitous in eukaryotes. Previous studies reported that PP2CD1 and SIPP2C38 efficiently catalysed dephosphorylation of AHA2 (H⁺-

ATPase) of the plasma membrane in Arabidopsis and tomatoes (Spartz *et al.*, 2014, 2017). The vacuolar H⁺-ATPase (V-ATPase) drove vacuolar acidification by transporting protons across the tonoplast into the vacuole to determine fruit acidity (Shiratake and Martinoia, 2007; Sweetman *et al.*, 2009; Hu *et al.*, 2016). We confirmed that MdPP2CH interacted with MdVHA-A3, MdVHA-B2, MdVHA-D2 and the vacuolar malate channel MdALMTII in the tonoplast and cytoplasm. The T268A variation in MdPP2CH led to a significant decline of MdPP2CH phosphatase activity, and this variation was responsible for the MdPP2CH-dependent variation in fruit malate (Figures 3 and 4). Our results are consistent with previous reports that natural variation at apple *MdALMTII* and soybean *PP2C-1* enhanced apple fruit acidity and soybean yield, respectively (Ma *et al.*, 2015b; Lu *et al.*, 2017).

The SAUR family comprises the largest number of auxin-induced genes in diverse plant species and regulates cellular, physiological and developmental processes (Ren and Gray, 2015). SAUR19 plays a key role in plasma membrane H⁺-ATPase activation by inhibiting protein phosphatases in the PP2CD subfamily in Arabidopsis (Spartz *et al.*, 2014). SAUR9 inhibited SIPP2C38-mediated dephosphorylation in tomatoes (Spartz *et al.*, 2017). Our results confirm that MdSAUR37 interacted with MdPP2CH in the nucleus and cytosol by reducing MdPP2CH activity by approximately 35% and subsequently activated fruit malate accumulation (Figure 7). A 36-bp insertion in the promoter region of *MdSAUR37* significantly enhanced the expression of *MdSAUR37*, resulting in increased fruit acidity. Similar variations have been demonstrated in *MdMYB10* and *SIALMT9*, regulating anthocyanin and malate accumulation in apples and tomatoes, respectively (Espley *et al.*, 2009; Ye *et al.*, 2017).

MYB transcription factors were reported to directly influence vacuolar transporters (Hu *et al.*, 2016, 2017). However, qtl08.1 was under the G' threshold value of BSA-seq in the 'Zisai Pearl' × 'Red Fuji' population because the *MdMYB44* A/G SNP (−1010) genotypes of the maternal parent 'Zisai Pearl' were G/G homologous. The PPVs were significantly correlated with the observed phenotype values when the genetic effect of *MdMYB44* was neglected in 'Zisai Pearl' × 'Red Fuji' (Figure S9).

In conclusion, by combining MapQTL and BSA-seq approaches, four major QTLs responsible for apple fruit acidity were mapped on chromosomes 8 and 16. The variations in three hierarchical epistatic genes, *MdSAUR37*, *MdPP2CH* and *MdALMTII*, from the corresponding QTLs were experimentally confirmed to influence fruit malate content. The PPVs were consistent with the observed phenotype values of fruit malate content. The results show that phenotype values can be practically predicted by the GVs of the genes, indicating good potential for application in GBS for fruit acidity.

EXPERIMENTAL PROCEDURES

Plant materials

We used 1800 and 2637 F₁ hybrids from *M. domestica* 'Jonathan' × 'Golden Delicious' and *Malus asiatica* 'Zisai Pearl' × *M. domestica* 'Red Fuji' as independent pedigree-based populations. The F₁ hybrid trees were planted at a density of 2.5 m × 0.5 m. A total of 92 *Malus* germplasm accessions were used as a natural segregating population for marker validation. Trees of germplasm accessions on Baleng Crab (*Malus robusta* Rehd.) seedling rootstock were planted at a row and spacing of 2.5 m × 0.5 m.

Six intact ripening apples were harvested from each tree for phenotyping fruit malate content every year, and fruit samples were collected at 30, 60, 90, 120 and 150 DAFB for gene expression analysis. Young leaf samples were collected from each tree in late spring. The apple fruit skins were peeled off, and the flesh was cut into small pieces. The fruit and leaf samples were immediately frozen in liquid nitrogen and stored at −78°C until use.

Apple calli induced from the flesh of the cultivar 'Orin' were subcultured every 3 weeks on Murashige and Skoog (MS) medium supplemented with 1.0 mg L^{−1} 2,4-dichlorophenoxyacetic acid (2,4-D) and 1.0 mg L^{−1} 6-benzylaminopurine (6-BA) at 25°C under dark conditions.

Phenotyping of fruit malate content

Fruit malate was extracted and separated by high-performance liquid chromatography (HPLC; Waters 600 chromatograph, <http://www.waters.com/>). The malate was quantified using a dual-λ ultraviolet detector (Waters 2487) as described by Sun *et al.* (2015).

MapQTL analysis

MapQTL 6.0 software (Van Ooijen, 2009) was used for map-based QTL analysis (Sun *et al.*, 2015; Tan *et al.*, 2017). We obtained phenotype data for 80, 122, 98 and 96 hybrids of 'Jonathan' × 'Golden Delicious' in 2011, 2014, 2015 and 2016, respectively, and 130 and 89 hybrids of 'Zisai Pearl' × 'Red Fuji' in 2015 and 2016, respectively. We detected QTLs using the Kruskal–Wallis (KW) test and interval mapping (IM) analysis. Markers near the QTLs were used as cofactors in the multiple QTL model (MQM) analysis for QTL detection. After a permutation test with 1000 replicates, logarithm of the odds (LOD) thresholds of 3.5 in 'Jonathan' × 'Golden Delicious' or 2.8 in 'Zisai Pearl' × 'Red Fuji' were used to find significant QTLs at the 95% confidence level.

BSA-seq analysis

Of the 'Jonathan' × 'Golden Delicious' and 'Zisai Pearl' × 'Red Fuji' F₁ hybrids, respectively 39 and 26 individuals with 3-year-stable highest fruit malate content and 37 and 33 individuals with the lowest fruit malate content were chosen for BSA-seq analysis. Genomic DNA of each fruit malate extreme was pooled to construct segregant bulks. Each parent or bulked DNA sample was sequenced over 30× the genome size using a paired-ends 150 strategy (Illumina X10, Illumina, <https://www.illumina.com/>).

The clean sequencing reads were mapped to the double haploid apple genome using Burrows–Wheeler alignment software (Li and Durbin, 2009; Daccord *et al.*, 2017). SAMtools software was used to obtain the read depth of the genome (Li *et al.*, 2009). A modified G-value method was used for the statistical analysis of allelic

variations between the two bulks (Magwene *et al.*, 2011). The pipeline and procedures of candidate gene mining will be published elsewhere soon.

RNA extraction, RT-PCR and qRT-PCR assay

Genomic DNA of young leaves and total RNA of apple flesh were isolated as previously described (Hu *et al.*, 2016). Complementary DNA was synthesized from total RNA using a cDNA Synthesis Kit (Takara, <http://www.takara-bio.com/>) and quantitative real-time PCR (qRT-PCR) was performed using the SYBR green reagent (Takara) in an Applied Biosystems 7500 real-time PCR system Thermo Fisher Scientific, (USA) as described by Xu *et al.* (2017). The threshold cycle (Ct) of each gene was investigated and normalized to the Ct value of the internal control gene *EF1α*. Quantitative RT-PCR was performed on three biological replicates containing three technical replicates each. The genes and primer sequences examined in the qRT-PCR experiments are listed in Table S9.

Plasmid construction and genetic transformation

To construct the *MdPP2CH* overexpression vector, the full-length CDS of *MdPP2CH* was isolated from 'Jonathan' and 'Golden Delicious' apples. The ORF of *MdPP2CH* was digested with *PacI* and *Ascl* and cloned into the pMDC83-GFP vector. To construct a RNAi vector, a 219-bp fragment (located between +535 bp and +753 bp in the full-length cDNA) of *MdPP2CH* was inserted into the *Ascl*/*NcoI* and *XbaI*/*PacI* sites of the pFGC5941 vector (Schwach *et al.*, 2005; Meng *et al.*, 2014). The recombinant plasmids (35S: *MdPP2CH*-GFP and RNAi:*MdPP2CH*) were introduced into 'Orin' apple calli as previously described (Horsch *et al.*, 1988; Hu *et al.*, 2016).

A 2672-bp promoter fragment of *MdPP2CH* and a 1400-bp promoter fragment of *MdSAUR37* were cloned from genomic DNA of 'Jonathan' and 'Golden Delicious', respectively. The promoter fragments were digested with *KpnI*/*NcoI* and cloned into the pCambia1301-GUS plant transformation vector. The resulting constructs, *MdPP2CH*Pro:GUS and *MdSAUR37*Pro:GUS, were transformed into leaf epidermal cells of 4-week-old *N. benthamiana* plants by infiltration with *Agrobacterium* strain GV3101. After transfection for 2–3 days, the transgenic leaves were dyed using a β -Galactosidase Reporter Gene Staining Kit (Solarbio, <http://en.solarbio.com/>) as described by Zhang *et al.* (2016). All primer sequences for vector construction are listed in Table S9.

Protoplast isolation and transformation

Protoplasts isolated from apple calli were prepared as described by Hu *et al.* (2016) and transformed with PEG using recombinant plasmid including *MdPP2CH*Pro:GUS and *MdSAUR37*Pro:GUS (He *et al.*, 2007; Yoo *et al.*, 2007). An empty vector pCambia1301 and 35S:GFP were used as the positive control and comparative control, respectively. After 16 h, the protoplasts were harvested and stored at -80°C until RNA extraction and GUS expression analysis (Zhang *et al.*, 2016). Three independent biological replicates with three technical replicates each were tested.

V-ATPase activity assay

Isolation of the tonoplast membranes assays was performed as described by Terrier *et al.* (2001), and the proton transport activity of V-ATPase was measured using a fluorescence spectrophotometer (Lu *et al.*, 2002; Hu *et al.*, 2017). Three independent biological replicates with three technical replicates each were tested.

Protein expression and phosphatase assay

The coding sequences of *MdPP2CH* and *MdSAUR37* were digested with *Bam*HI/*Sma*I and cloned into a pGEX-4T-1 vector for protein expression in BL21 (DE3) *Escherichia coli* cells. Purification of GST-tagged recombinant proteins was performed as described by Spartz *et al.* (2014). All primer sequences for vector construction are listed in Table S9.

The phosphatase activity was measured using pNPP as a substrate (Dupeux *et al.*, 2011; Spartz *et al.*, 2014). For the pNPP assay, a 100- μl reaction containing 25 mM TRIS-HCl (pH 7.5), 2 mM MnCl_2 , 1 mM dithiothreitol (DTT), 5 mM pNPP and the indicated PP2C was performed. The absorbance at 405 nm was recorded every minute for 20–30 min using a Bio-Rad 680 instruments (<http://www.bio-rad.com/>). Three independent biological replicates with three technical replicates each were tested.

Subcellular localization

The coding sequences of *MdPP2CH* and *MdSAUR37* without the stop codon were amplified and cloned into the pMDC83:GFP expression vector. The fusion constructs were transformed into leaf epidermal cells of 4-week-old *N. benthamiana* or protoplasts of apple calli. The fluorescence was detected 2–3 days following transfection by confocal microscopy. 4',6-Diamidino-2-phenylindole (DAPI; Invitrogen, <http://www.invitrogen.com/>) was used to stain the nucleus (Hu *et al.*, 2017). *N*-(3-Triethylammoniumpropyl)-4-(6-(4-(diethylamino)phenyl)hexatrienyl)pyridinium dibromide (FM4-64; Invitrogen) was used to stain the tonoplast (Kataoka *et al.*, 2004; Gattolin *et al.*, 2011).

Protein-protein interaction assay

A Y2H assay was performed as described by Hu *et al.* (2016). The full length ORF of *MdPP2CH* was cloned into pGADT7 and *MdSAUR37*, *MdVHA-A3*, *MdVHA-B2*, *MdVHA-D2* and *MdALMT7* were cloned into pGBKT7. The plasmids with AD and BD were co-transformed into Y2H Gold yeast (Clontech, <http://www.clontech.com/>). For interaction screening, yeast cultures were grown on –Leu/–Trp medium for the transformation control, and the colonies were transferred into –Leu/–Trp/–His/–Ade medium with X- α -Gal for the interaction studies.

The BiFC assay was conducted to validate protein–protein interactions (Walter *et al.*, 2004). The CDS without the stop codon of *MdPP2CH* was cloned into pSPYNE-35S and *MdSAUR37*, *MdVHA-A3*, *MdVHA-B2*, *MdVHA-D2* and *MdALMT7* were cloned into pSPYCE-35S. The fusion plasmids were co-transformed into leaf epidermal cells of *N. benthamiana* or protoplasts of apple calli (Schütze *et al.*, 2009). The transformed leaf epidermal cells and protoplasts were observed using confocal microscopy.

Statistical analysis

A generalized mixed linear model was used to analyse the associations between molecular markers and malate content using TASSEL software, version 4.0 (Yu *et al.*, 2006; Bradbury *et al.*, 2007). Differences between the control and experimental variants/treatments were analysed by one-way analysis of variance (ANOVA) through Dunnett's multiple comparison at a significance level of $\alpha = 0.05$.

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AUTHOR CONTRIBUTIONS

XZ and ZH conceived of and designed the experiments. DJ and FS conducted the experiments. YW, TW, XX and ZH contributed plant materials, phenotyped apple fruit malate content or conducted statistical analysis. DJ and XZ wrote the manuscript. All authors read and approved the manuscript.

CONFLICT OF INTERESTS

The authors declare no competing financial interests.

SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article.

Figure S1. Phylogenetic analysis of Arabidopsis type-2C protein phosphatases and MDP0000141481 (MdPP2C) using MEGA 6. Numbers indicate bootstrap support based on 1000 replicates.

Figure S2. Amino acid sequence alignment of MdPP2CH was cloned from 'Jonathan' (J) and 'Golden Delicious' (G).

Figure S3. Western blotting analysis of recombinant bacterial GST-tagged fusion protein of 268T and 268A variants in the MdPP2CH protein using anti-GST antibody.

Figure S4. Western blotting analysis of MdPP2CH-GFP protein in wild-type and 35S:MdPP2CH-GFP transgenic apple calli using anti-GFP antibody. The ACTIN protein was used as a loading control.

Figure S5. Comparison of the 184-bp deletion and promoters of MdPP2CH cloned from 'Jonathan' (J) and 'Golden Delicious' (G).

Figure S6. Phylogenetic analysis of the Arabidopsis SAUR protein and MDP0000153382 (MdSAUR) using MEGA 6. Numbers indicate bootstrap support based on 1000 replicates.

Figure S7. Comparison of the MdSAUR37 promoters cloned from 'Jonathan' (J) and 'Golden Delicious' (G).

Figure S8. Western blotting analysis of recombinant bacterial GST-tagged fusion protein of MdSAUR37 using anti-GST antibody 'Jonathan' (J) and 'Golden Delicious' (G).

Figure S9. Association between genotypes of MdSAUR37, MdPP2CH, and MdALMTII and fruit malate content in apple hybrids and germplasm accessions.

Figure S10. Proposed model for the role of MdSAUR37, MdPP2CH, MdALMTII and MdMYB44 in regulating the malate content of apple fruit.

Table S1. Candidate genes searched from the regions of quantitative trait loci and the association between variations and malate content in individuals with extremely high and low malate content.

Table S2. The linkage analysis between single-nucleotide polymorphism A/G (2831) and the 184-bp deletion (2901) in MdPP2CH was validated by sequencing 30 *Malus* germplasm accessions.

Table S3. The linkage analysis between single-nucleotide polymorphism (SNP) A/G (−1010) and SNP T/− (−536) in MdMYB44 was validated by sequencing 30 *Malus* germplasm accessions.

Table S4. The linkage analysis of MdMYB44, MdPP2CH, MdSAUR37 and MdALMTII genotypes in 246 and 123 randomly chosen hybrids of 'Jonathan' × 'Golden Delicious' and 'Zisai Pearl' × 'Red Fuji', respectively, and 92 *Malus* accessions.

Table S5. The recombination rates between MdMYB44, MdPP2CH and MdSAUR37 in 246 and 123 randomly chosen hybrids of 'Jonathan' × 'Golden Delicious' and 'Zisai Pearl' × 'Red Fuji', respectively, and 92 *Malus* accessions (%).

Table S6. The genotypes of MdSAUR37, MdPP2CH, MdALMTII, MdMYB44 and fruit malate content in 246 and 123 randomly chosen F₁ hybrids of 'Jonathan (J)' × 'Golden Delicious (G)' and 'Zisai Pearl (Z)' × 'Red Fuji (F)', respectively, and 92 *Malus* accessions.

Table S7. Calculated genotype values assigned to four genes MdSAUR37, MdPP2CH, MdALMTII and MdMYB44.

Table S8. The combinative genotypes and predicted phenotype values were analysed in 246 and 123 randomly chosen F₁ hybrids of 'Jonathan (J)' × 'Golden Delicious (G)' and 'Zisai Pearl (Z)' × 'Red Fuji (F)', respectively.

Table S9. The primers used for PCR and qRT-PCR.

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